Creative Workspace App

Complete Feature Inventory — Version 1.0

Project Status: Brainstorming / Predevelopment
Platform: PC Desktop
Development Approach: AI-assisted coding with support from an experienced developer
Initial Purpose: Personal use, with optional collaborative workspaces
Scope: Full feature vision, not the initial MVP

---

1. Project Overview

The goal is to create a desktop creative workspace inspired by Milanote, designed primarily for artists, illustrators, comic creators, manga creators, writers, and other visual storytellers.

The application will combine infinite visual canvases, interconnected boards, notes, reference images, document viewing, artwork management, and project organization into one flexible environment.

It should support the creative process from brainstorming and research through planning, reviewing, and organizing finished work.

Core Design Principles

- **Creative freedom:** Users should not be forced into predefined organizational structures.
- **Art-focused but not manga-exclusive:** All features should be available in every project.
- **Visual organization first:** Infinite canvases are the foundation of the application.
- **Optional structure:** Tools such as grids, guides, tags, and templates should assist users without becoming mandatory.
- **Original file preservation:** The application should organize and display artwork without altering the original files.
- **Review over creation:** Dedicated art and writing programs will handle most content creation.
- **Expandable design:** The application should be structured to accommodate additional features later.
- **No integrated AI:** AI may assist with developing the application, but no live AI functionality is planned inside it.

The application is not intended to replace Clip Studio Paint, Procreate, Photoshop, Microsoft Word, or professional drawing and writing software.

---

2. Platform and Multi-Window Support

Desktop Application

- PC-only application.
- Optimized for keyboard and mouse interaction.
- No mobile or tablet application required.
- Support standard desktop window resizing and fullscreen use.

Multiple Windows

Users should be able to open multiple windows of the same application.

Features:

- Open separate boards in different windows.
- Access the same project from multiple windows.
- Automatically synchronize saved changes between windows.
- Independently navigate, zoom, and scroll in each window.
- Use multiple windows regardless of the number of monitors.
- Support dual-monitor setups without requiring any special configuration.

Multi-window functionality should remain relatively simple.

Advanced docking, complex window management, and dragging tabs between windows are unnecessary.

---

3. Projects and Organizational Structure

Independent Projects

Each creative project should function as its own separate workspace.

Each project can contain:

- A main project board.
- Unlimited-style nested boards.
- Project-specific folders.
- An asset library.
- Documents and notes.
- Artwork and reference images.
- Collections.
- Tags and navigation tools.

Projects should remain independent rather than sharing one universal asset library.

Folders and Nested Boards

Both organizational methods should exist.

Traditional folders:

- Organize files and materials through a folder hierarchy.
- Access folders through a sidebar or file explorer.
- Support nested folders.

Visual nested boards:

- Place boards inside other boards.
- Open nested boards directly from the canvas.
- Allow multiple levels of nesting.
- Navigate through the hierarchy without relying exclusively on folders.

Users should be able to organize projects visually or through a traditional file-management system.

Collections

Collections are distinct from ordinary folders.

They organize groups of content with additional viewing functionality.

Examples include:

- Manga chapters.
- Page sequences.
- Artwork galleries.
- Reference collections.

A collection can appear as an object on a board.

---

4. Infinite Canvas

The infinite canvas is the primary application interface.

Canvas Navigation

- Effectively unlimited horizontal and vertical space.
- Free panning in all directions.
- Vertical and horizontal scrolling.
- Smooth zooming in and out.
- Mouse and keyboard navigation.
- Navigation between nested boards.
- Ability to fit board contents into the visible screen area.

The canvas should not have traditional page boundaries.

Object Manipulation

Board objects should support relevant basic interactions:

- Dragging and repositioning.
- Resizing.
- Selecting multiple objects.
- Copying and duplicating.
- Grouping and ungrouping.
- Locking objects in place.
- Adjusting object layering.
- Aligning objects.
- Moving grouped objects together.

Text notes and comments should support free rotation.

Freeform and Structured Layouts

Freeform placement should be the default.

Optional tools should include:

- Visible grid guidelines.
- Snap-to-grid.
- Alignment guides.
- Object alignment.
- Equal spacing and distribution.
- Uniform object sizing.

Displaying guidelines should not automatically reposition objects.

Users should be able to create either completely freeform arrangements or highly organized, uniform layouts.

---

5. Board Object Types

The canvas should support multiple object types.

Text Boxes and Notes

- Freely placed text boxes.
- Resizable text areas.
- Basic text formatting.
- Adjustable font sizes.
- Text colors and backgrounds.
- Rotatable notes.
- Short-form writing directly on the canvas.
- Ability to arrange reference images around text.
- Support images inside text cards where practical.

These are intended for character biographies, setting descriptions, worldbuilding notes, brainstorming, and similar material.

Images

- Place images directly on boards.
- Move, resize, and rotate images.
- Preserve original image proportions.
- Preserve transparency.
- Open images in a full-size viewer.
- Link images to their original asset files.

Documents

- Place documents on boards as file objects.
- Open documents in a dedicated viewer.
- Display documents using compact cards, covers, or previews.

Nested Boards

- Represent other boards as clickable objects.
- Support custom covers.
- Open the linked board when activated.

Checklists

- Create editable checklists.
- Add and remove checklist entries.
- Mark individual items as complete.
- Use checklists as standalone canvas objects.

Tables

- Create simple tables directly on boards.
- Add or remove rows and columns.
- Resize tables.
- Edit cell contents.
- Apply basic formatting.

Advanced spreadsheet functionality is not required.

Headers and Dividers

- Create section headings.
- Add visual dividers between board areas.
- Resize and reposition headers and dividers.
- Use them to organize large canvases.

Containers

- Create visual containers around related objects.
- Move related items together.
- Apply basic container styling.

- Use containers to organize reference groups, character materials, and other visual sections.

Connections and Links

- Connect objects using lines or arrows.
- Link objects to boards, documents, or other materials.
- Add website links and link cards.

Comments

- Add free-floating comments.
- Move comments anywhere on a board.
- Rotate and resize comments.
- Attach comments to specific objects when needed.

---

6. Document Viewing and Review

The application should support large writing files without attempting to become a complete writing program.

It should comfortably accommodate substantial documents, including manuscripts or scripts hundreds of pages long.

Document Viewer Features

- Open supported writing documents and PDFs.
- Read lengthy documents.
- Search for specific words or phrases.
- Navigate through a linked table of contents.
- Jump to chapters, headings, scenes, or sections.
- Create custom bookmarks and navigation points.
- Add comments and review notes.
- Make limited text corrections where the file format permits.
- Download the original document.

The application should prioritize reviewing and navigating documents rather than extensive editing or formatting.

A 500-page manuscript should be a dedicated document, not one enormous canvas text box.

---

7. Artwork and Image Handling

Image Preservation

Images must never be distorted during resizing.

Requirements:

- Maintain original aspect ratios.
- Scale images proportionally.
- Preserve original image quality and resolution in storage.
- Preserve transparent backgrounds.
- Avoid automatic cropping.
- Do not reshape images.
- Do not modify original artwork files.

The application is not intended to provide image-editing tools.

Full-Size Image Preview

Users should be able to open supported images in a larger viewer.

Features:

- Full-size viewing.
- Zooming.
- Panning.
- Fit-to-window viewing.
- Navigation between images in supported collections.
- Original-file downloading.

Native Artwork File Support

The app should accept and preserve a broad range of creative file formats.

Target formats include:

- PNG

- JPEG/JPG
- WebP
- TIFF
- PSD
- PDF
- Clip Studio Paint files (.clip)
- Procreate files (.procreate)
- Other relevant artwork and export formats.

Related multi-file project formats should be considered where necessary.

Important distinction: Storing a file does not necessarily mean the application can render it.

For formats that cannot be automatically previewed, users should be able to attach a standard image preview while keeping the original working file available for download.

The goal is broad file compatibility and preservation, not native editing support.

---

8. Project Asset Libraries

Each project should have its own asset library.

Asset Storage

- Store imported images, documents, and other files.
- Preserve original files.
- Maintain file metadata.
- Organize assets into folders.
- Search and filter stored assets.
- Track file sizes and overall storage usage.

Reusable Assets

The same file should be usable on multiple boards without duplicating its original storage.

For example, a character design could appear on a character board, chapter board, and costume board while referencing one underlying asset.

Requirements:

- One original asset can have multiple board appearances.

- Each appearance can have its own position and display settings.
- Removing an appearance does not delete the original.
- Moving a file between library folders should not break references.
- Deleting an original file should warn users if it is still referenced elsewhere.

Mass Importing

Users should be able to import large numbers of assets directly into the project's library.

Features:

- Multiple-file selection.
- Drag-and-drop importing.
- Importing entire folders.
- Preserving folder structures where possible.
- Choosing a destination asset folder.
- Bulk organization and tagging.

Mass importing should not require placing every file onto a board.

Storage Approach

Both local and cloud storage remain under consideration.

The storage system should support personal projects and eventual shared workspaces while keeping costs manageable.

The specific storage provider and local/cloud architecture have not yet been selected.

---

9. Image Stacks

Image stacks are a visual grouping feature for images on a board.

They are not separate assets in the project library.

Stack Features

- Select multiple images and combine them into one stack.
- Display one image at a time.
- Click through images with navigation arrows.

- Reorder images inside the stack.
- Choose a default cover image.
- Expand the stack into a slideshow viewer.
- Separate a stack back into individual images.
- Download individual original images.

Possible uses:

- Sketch, inked, and finished artwork.
- Different character outfits.
- Character designs at different ages.
- Alternate design concepts.
- Different character expressions.
- Related visual references.

The underlying files remain independent assets.

---

10. Page Collections and Manga Planning

Page collections should organize sequential artwork.

These tools must remain available to all projects, not just manga projects.

Page Collections

- Combine images into ordered page sequences.
- Rearrange pages.
- Add or remove pages.
- Choose collection covers.
- Navigate directly to individual pages.
- Organize pages into chapters or other sequences.

Page Planner

- Display page thumbnails in order.
- Create empty page placeholders.
- Reserve pages before artwork is finished.
- Rearrange planned page order.
- View pages as single pages or spreads.
- Account for right-to-left and left-to-right reading.

- Plan double-page spreads.
- Plan important page turns and reveals.
- Track page counts.

Page Replacement

- Replace rough artwork with updated versions.
- Preserve the page's position in its collection.
- Optionally preserve earlier artwork versions.
- Keep links and related organizational information intact where possible.

Reader and Presentation Mode

Page collections should have a full-screen viewing mode similar to PowerPoint's presentation mode.

Features:

- Full-screen reading.
- Single-page viewing.
- Two-page spread viewing.
- Continuous vertical viewing.
- Page-by-page navigation.
- Thumbnail navigation.
- Zooming and panning.
- Configurable reading direction.
- Clean presentation without editing controls.

A separate review experience should allow comments and annotations to remain visible when needed.

---

11. Artwork Annotation and Review

The application should provide only lightweight annotation tools.

Annotation Features

- Highlighting.
- Circling areas.

- Drawing simple lines and arrows.
- Adding text comments.
- Floating notes.
- Movable and rotatable comments.
- Repositioning annotations.
- Hiding or showing annotation layers.
- Removing annotations without affecting the original file.

Annotations should be non-destructive.

The application does not need painting brushes, artistic layer systems, advanced selections, or any substantial drawing functionality.

Revision Support

- Preserve previous artwork versions when desired.
- Replace displayed artwork with updated versions.
- Review different stages of the same artwork.
- Keep review notes associated with the appropriate version where practical.

---

12. Tags and Advanced Searching

Tags should handle much of the application's character, artwork, and production tracking.

There should be no mandatory specialized character-tracking database.

Two Types of Tags

Unique tags:

- Created for a particular item or limited purpose.
- Do not need to become permanent reusable tags.

Reusable tags:

- Stored in a project's tag library.
- Available across the project.
- Applicable to multiple files and objects.

Tag Management

- Create tags.
- Rename tags.
- Remove tags.
- Apply multiple tags to an object.
- Apply tags to multiple items simultaneously.
- Use tag autocomplete.
- Use tags on artwork, documents, boards, pages, and other appropriate items.

Advanced Tag Search

- Search by one or more tags.
- Match all selected tags.
- Match any selected tag.
- Combine tag filters.
- Filter by file or object type.
- Search across the entire project.
- Quickly locate all matching assets or objects.

Possible tags include character names, chapters, production stages, locations, themes, and review statuses.

Tags should remain customizable rather than forcing a predefined production workflow.

---

13. Linked Tables of Contents and Navigation

The application should have a strong linked navigation system.

Hybrid Table of Contents

Tables of contents should support both automatic and manual organization.

Features:

- Generate entries from headings or other relevant objects.
- Manually create entries.
- Edit automatically generated entries.
- Rearrange entries.
- Remove unwanted entries.

- Customize labels.

Supported Link Destinations

A table of contents should be able to link to:

- Locations on an infinite board.
- Named canvas anchors.
- Individual objects.
- Nested boards.
- Documents.
- Document headings or bookmarks.
- Page collections.
- Specific pages where supported.

Clicking a linked board destination should navigate directly to the appropriate content.

Named Anchors

Users should be able to designate and name specific locations on a canvas.

These locations can then be linked from a TOC or other navigation object.

Navigation Hierarchy

The application should also support a conventional folder/sidebar hierarchy for navigating projects and nested boards.

The linked TOC is intended to reduce the need to manually navigate enormous canvases or automatically restore every previously visited canvas position.

---

14. Pinned Items and Workspace Navigation

The interface should include a section for pinned content.

Pinned Shortcuts

Users should be able to pin frequently used materials, including:

- Tables of contents.
- Boards.
- Documents.
- Images.
- Page collections.
- Specific board locations.
- Other active project materials.

Pinned shortcuts should allow quick access without moving or duplicating the original object.

Pinned Content Panels

The interface should support keeping useful material accessible while working elsewhere.

Examples:

- A pinned table of contents.
- A reference image.
- A checklist.
- An active document preview.

The pinning area should be collapsible to preserve canvas space.

The goal is quick navigation and access to frequently used materials rather than complicated window management.

---

15. Collaboration and Shared Workspaces

Projects should initially be private, with optional sharing.

Shared Projects

- Invite collaborators to specific projects.
- Keep unrelated projects private.
- Allow collaborators to view project materials.
- Support editing and commenting permissions.
- Share boards, documents, artwork, and organizational information within the project.

Single-Editor Collaboration

Simultaneous editing of the same item is not required.

A simpler editing model is preferred:

- One person edits a particular object or document at a time.
- Other collaborators can view it.
- Editing access becomes available when the current editor finishes.
- Saved changes become available to other collaborators.
- Different people may work on different items within the same project.

The system should prevent conflicting edits and handle interrupted editing sessions.

Live cursors, real-time coauthoring, and Google Docs-style simultaneous editing are unnecessary.

---

16. Plot Roadmaps

Priority: Lower / Optional

A visual plot-roadmap feature should remain under consideration.

It would allow users to organize narrative developments through connected paths.

Potential functionality:

- Plot-event cards.
- Multiple story paths.
- Branching storylines.
- Converging storylines.
- Merging narrative threads.
- Linking plot events to other materials.
- Rearranging connected events visually.

The roadmap may be built using existing board objects and connection tools rather than requiring a completely separate system.

This feature is desirable but not essential.

---

17. Covers, Typography, and Appearance

Custom Covers

Support custom cover artwork for:

- Boards.
- Nested boards.
- Documents.
- Files.
- Collections.

Users should be able to choose an image that represents the content without modifying the original file.

Font Support

Explore support for importing custom fonts.

Desired capabilities:

- Import common desktop font formats such as TTF and OTF.
- Use imported fonts in compatible text objects.
- Apply custom typography to relevant board content.
- Preserve appropriate font behavior when opening shared projects.

Font licensing and portability will need consideration during development.

---

18. Quick Capture and Importing

The application should make collecting creative references easy.

Clipboard and Drag-and-Drop

- Copy images from a browser and paste directly onto a canvas.
- Drag images from Windows File Explorer onto a board.
- Drag multiple files into the application.
- Paste URLs to create link cards.
- Import files without manually navigating complex menus.

Imported files should be registered with the appropriate project asset library.

Direct Library Import

Users should also be able to import files directly into folders in their asset library, without creating canvas objects.

---

19. Keyboard Shortcuts and Quick Commands

The PC application should support efficient keyboard-driven navigation.

Keyboard Shortcuts

Provide shortcuts for common actions such as:

- Creating notes.
- Copying and pasting.
- Duplicating objects.
- Grouping objects.
- Undo and redo.
- Searching.
- Opening boards.
- Navigating between content.
- Other frequently used actions.

Command Search

Include a searchable command menu for quickly finding and executing available actions.

This should reduce the need to navigate menus for common operations.

---

20. Board History and Recovery

Users should be able to recover from mistakes and unwanted changes.

Undo and Redo

- Undo recent actions.
- Redo undone actions.
- Support common board-editing operations.

Board Snapshots

- Save snapshots of a board's arrangement.
- Restore previous arrangements.
- Preserve recovery points before significant reorganizations.

Deleted Content Recovery

- Deleted objects should go to a recoverable trash area when appropriate.
- Support restoration of accidentally deleted content.
- Avoid permanently deleting underlying assets merely because their board appearances were removed.

---

21. Essential Quality-of-Life Features

These features should be considered fundamental usability requirements.

Saving and Reliability

- Automatic saving.
- Visible save status.
- Protection against accidental data loss.
- Reliable recovery mechanisms.
- Backup planning.

Standard Editing Actions

- Copy.
- Cut.
- Paste.
- Duplicate.
- Delete.

- Undo.
- Redo.
- Multi-select.
- Bulk actions.

Trash and Recovery

- Recycle bin or trash system.
- Restore deleted content.
- Clear distinction between removing a board object and deleting its underlying file.

Storage Information

- File-size warnings.
- Project storage usage information.
- Visibility into large files and storage consumption.

Accessibility and Navigation

- Keyboard navigation.
- Reasonable accessibility support.
- Clear interaction feedback.
- Searchable interface commands.

Project-Wide Search

Search should work across relevant content within a project, including:

- Board names.
- Text notes.
- File names.
- Documents where text is searchable.
- Tags.
- Other indexed metadata.

Search results should allow users to navigate to the matching content.

---

22. Features Explicitly Outside the Planned Scope

The following are not currently part of the application's intended functionality:

- A mobile application.
- A full illustration or painting program.
- Advanced image editing.
- Cropping or reshaping artwork.
- A full Microsoft Word-style document editor.
- Live AI assistants or AI generation features.
- Simultaneous collaborative editing of the same document or object.
- Advanced multi-window docking and window-management systems.
- Mandatory manga-specific project templates.
- Mandatory character databases or production-tracking systems.

The goal is to avoid duplicating functions better handled by dedicated creative software.

---

23. Decisions Still to Be Made

These are implementation or design choices that have not yet been finalized:

- Programming languages and frameworks.
- Desktop application framework.
- Database architecture.
- Local, cloud, or hybrid file storage.
- Cloud storage provider.
- How shared projects are hosted.
- Backup and recovery implementation.
- Supported document formats and limitations.
- Automatic preview generation for specialized art formats.
- Custom font handling.
- Detailed interface styling.
- Exact project and folder navigation interface.
- Storage limits and anticipated operating costs.
- Performance strategies for very large boards.
- Feature development order.
- Initial MVP scope.

These should be investigated with the developer before implementation decisions are finalized.

---

24. Development Philosophy and Next Steps

This document represents the full feature vision, not a commitment to build everything immediately.

The planned development process is:

Stage 1 — Feature Inventory

Collect and document desired features without prematurely limiting ideas.

Stage 2 — Feasibility Review

Evaluate each feature based on:

- Technical complexity.
- Development effort.
- Implementation cost.
- Ongoing operating cost.
- Dependencies on other features.
- Importance to the overall experience.

Stage 3 — Scope Selection

Classify features as:

- Essential for the first working version.
- Important for later development.
- Optional enhancements.
- Too complex or impractical for available resources.

Stage 4 — Architecture Planning

Choose the software architecture and technologies needed to build a maintainable application.

Stage 5 — MVP Development

Build a functional initial version centered on the application's most important systems, then expand incrementally.

The overarching goal is a **flexible, PC-based visual creative workspace that can organize an entire artistic project, from early ideas and reference gathering to scripts, finished artwork, and collaborative review, without requiring users to adopt a rigid workflow.**